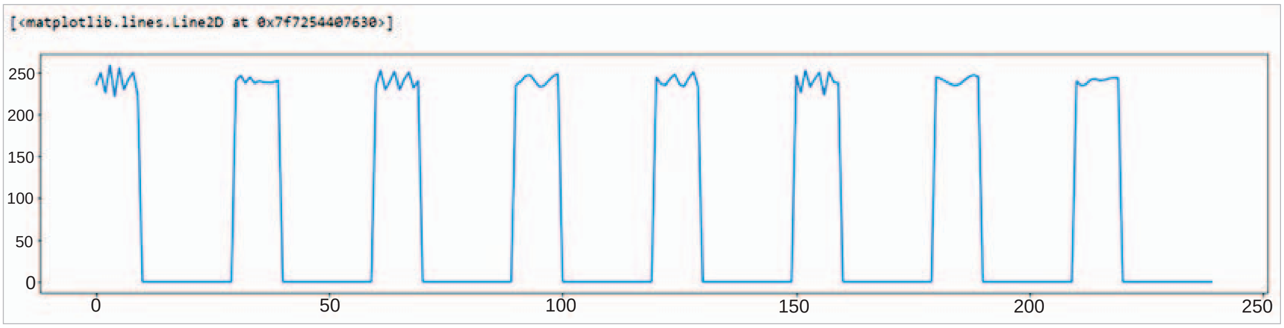

Figure 3: DTMF frequency transformation

Hence, we execute the decoding code in the *dtmf_decode()* function by calling the functions given below:

```
print(decode_dtmf (tone_2)) # works
perfectly!!
['7', '6', '5', '8', '9', '4', '1', '0']
print(decode_dtmf(tone_1))
['1', '4', '9', '8', '*', '0']
```

Now, we can see how *tone_1* and *tone_2*, which were an overlapping sinusoidal wave of frequency '2', are being converted to the original sequence with minimal error.

Figure 3 shows the energy graph of DTMF frequency transformation.

DTMF is used even today. Often, when dialling a customer care number, we are greeted with the automated response of 'For English, press 1, for Hindi press 2', and so on. This selection menu on voice calls would have been a challenge were it not for DTMF. The key that the user taps for a particular selection is encoded as DTMF and then decoded at the receiver's end to get the user's choice.

However, the use of DTMF was not as simple as it seems. It was vulnerable to an attack called 'phreaking'. This attack used a malicious actor to send unauthorised DTMF signals by hooking up some custom hardware on the telephone lines. In some regions of the United States, it is illegal to connect anything to the telephone line for the same reason.

A lot of applications and systems can be built using the concepts explained in this article. So this retro technology is evergreen! END

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